



Engineering Professional Practice (EX40003)
7th Semester-Autumn 2026
Assignment-I

Answer the following questions

Due: 5th Aug. 2026

1. Differentiate between engineering as a discipline and engineering as a profession.
2. List any four essential attributes of a successful practicing engineer.
3. Explain why ethical responsibility is important in the engineering profession
4. Discuss the essential attributes of a practicing engineer and explain how these attributes influence their professional responsibilities.
5. Discuss the various functions of a practicing engineer during the planning, design, execution, and maintenance stages of an engineering project.
6. An engineer is a problem solver, designer, and change agent." Justify this statement with appropriate real-life engineering examples.
8. How can engineers contribute to achieving sustainable development and improving the quality of life?
9. Compare the roles of a mechanical engineer, civil engineer, and software engineer in solving societal problems.
10. Discuss how creativity, teamwork, and communication skills influence the success of an engineering project.
11. Explain the importance of lifelong learning for engineers in the era of Industry 4.0 and Artificial Intelligence.
12. Suggest five ways in which engineers can act as change agents for sustainable and inclusive development.
13. Explain the importance of lifelong learning for engineers in the era of Industry 4.0 and Artificial Intelligence.
14. Explain the impact of Artificial Intelligence on modern engineering practices with suitable examples.
15. Why should engineers continuously upgrade their technical knowledge and professional skills? Explain with examples.
16. Explain how emerging technologies such as AI, IoT, robotics, and digital twins are changing engineering design and manufacturing.

17. Explain the role of innovation and lifelong learning in solving future engineering challenges.
18. Why are interdisciplinary knowledge and teamwork becoming increasingly important for engineers in the digital age?
19. A manufacturing company experienced low productivity, frequent machine failures, and high production costs. Engineers introduced an **AI-based predictive maintenance system** that uses sensors and machine learning to monitor equipment health and predict failures before they occur.

Based on the above case study, answer the following:

- a. Identify the engineering problem.
- b. Explain the design approach used to solve the problem.
- c. Describe how the innovation has brought positive change to society.
- d. Mention the role of engineers in developing and implementing the solution.

20. A rapidly growing city faces heavy traffic congestion, long travel times, and frequent road accidents. Engineers designed a **smart traffic management system** using AI-based traffic signal control, CCTV cameras, and real-time traffic monitoring to optimize vehicle movement and reduce congestion.

Based on the above case study, answer the following:

- a. Identify the engineering problem.
- b. Explain the design approach used to solve the problem.
- c. Describe how the innovation has brought positive change to society.
- d. Mention the role of engineers in developing and implementing the solution.

Instruction for Students:

Students whose roll number ends with an odd digit must answer the odd-numbered questions (1, 3, 5, 7, ...).

Students whose roll number ends with an even digit must answer the even-numbered questions (2, 4, 6, 8, ...).

Course Coordinator

Dr. Pruthwiraj Sahu

Ph.-8249839827